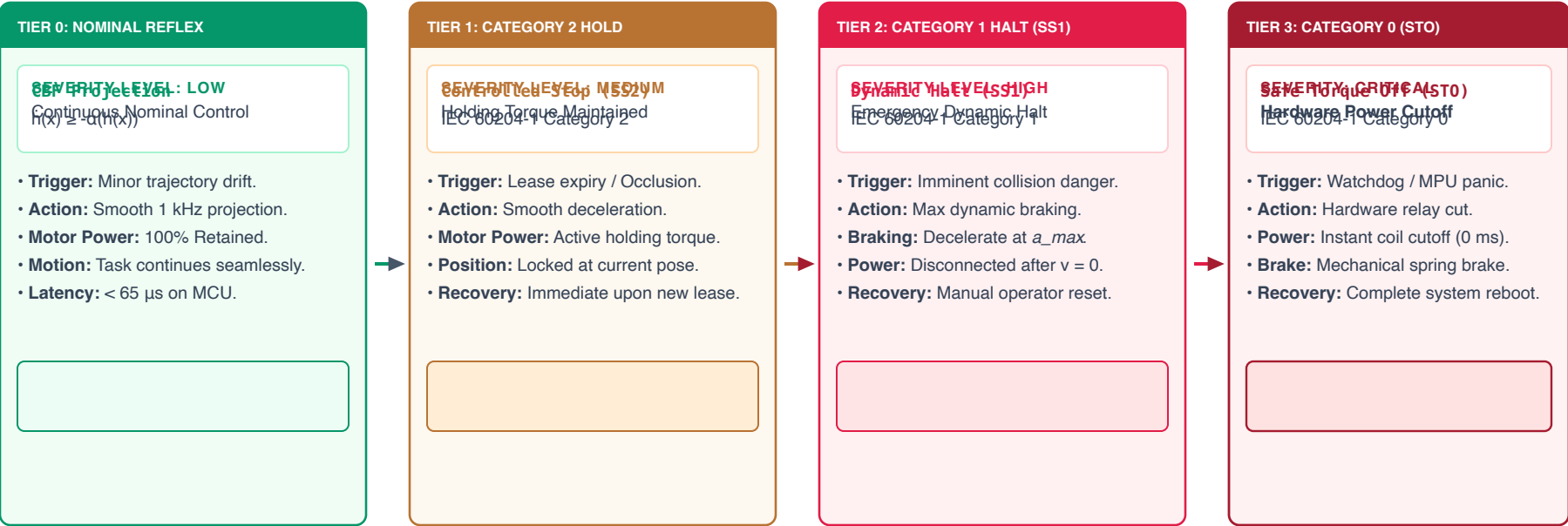


IEC 60204-1 STANDARDIZED EMERGENCY STOP & HARDWARE INTERLOCK HIERARCHY

Deterministic escalation tiers: from nominal CBF projection to Cat 2 position holds, Cat 1 dynamic halts, and Cat 0 Safe Torque Off (STO)



THE HARDWARE INTERLOCK & WATCHDOG INVARIANT

1. **Hardware Watchdog Interlock:** If the Linux MPU panics, crashes, or stalls for > 50 ms, the hardware watchdog pin transitions low.
2. **MCU Autonomy:** The real-time MCU holds independent gate driver authority and instantly trips Category 0 STO without software cooperation.
3. **Physical Containment:** Kinetic energy is absorbed safely by mechanical brakes, proving that software failures cannot produce catastrophic physical injury.